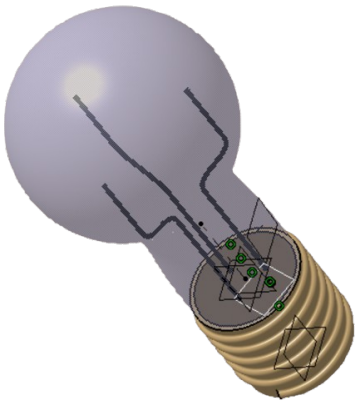
School of Air and Space Engineering

Parameterization of a Light Bulb Assembly

CATIA V5R19



Module: PARAMETERIZATION IN CATIA
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Introduction

This project is part of the **Parameterization in CATIA V5** module at IPSA. The primary objective of this practical work (TD) is to transition from traditional static CAD modeling to **Knowledge-Based Engineering (KBE)** by designing a smart, adaptive light bulb assembly.

In modern aerospace and systems engineering, the ability to automate design modifications is highly valuable. Rather than manually rebuilding geometry for every iteration, we embed "intelligence" directly into the 3D model. This is achieved using the **Knowledge Advisor** workbench in CATIA, which allows the definition of a logical framework governed by parameters, mathematical formulas, and conditional rules.

Throughout this technical report, we focus on applying KBE concepts through two structured exercises covering:

- **Part Parameterization**: Activating and deactivating structural features based on user-defined String parameters.
- **Assembly Automation**: Linking bulb dimensions to internal components using mathematical rules and formulas.
- **Quality Control & Catalogs**: Implementing Checks to monitor critical dimension ratios and using Design Tables to efficiently manage standard industrial configurations.

1 Exercise 1: Part Parameterization

The first exercise focuses on configuring the CATIA environment properly and creating basic logical rules at the Part level to control geometric features automatically.

1.1 Environment Settings

To enable full visibility of the design's embedded "intelligence", specific settings were modified under **Tools** > **Options**. The *Parameters and Measure* settings were updated to display parameters alongside their current values and formulas directly in the specification tree. Additionally, the display of Parameters and Relations was activated in the Infrastructure settings to allow real-time tracking of our variables.

1.2 User Parameters Setup

A new User Parameter of type *String* was created and renamed to `Type`. Using the "Add Multiple Values" function, two distinct string values were assigned to this parameter: "American" and "European". This discrete parameter acts as the main logical switch for the part's overall geometry.

1.3 Knowledge Rules Implementation

To automate the part's configuration without manual redesign, an *If-Then-Else* rule was written using the Knowledge Advisor workbench. The logic dictates that:

- If `Type == "AMERICAIN"`, the screw thread features are activated (`Activity = true`), while the pin features are deactivated (`Activity = false`).
- If `Type == "EUROPEEN"`, the logic is strictly reversed: the screw thread is deactivated, and the pin geometry becomes visible.

```
/*Règle créé(e) par vinterp 13/02/2024*/
if Type == "AMERICAIN"
{
        `Corps principal\Extrusion.1\Activité` = false
        `Corps principal\Symétrie.1\Activité` = false
        `Corps principal\Congé arête.9\Activité` = false
        `Corps principal\Nervure.2\Esquisse.4\Activité` =true
        `Corps principal\Hélice.2\Activité`= true
        `Corps principal\Nervure.2\Activité` = true
        `Corps principal\Congé arête.6\Activité` = true
        `Corps principal\Plan.2\Activité` = true
}
else
{
        `Corps principal\Hélice.2\Activité`= false
        `Corps principal\Nervure.2\Activité` = false
        `Corps principal\Nervure.2\Esquisse.4\Activité` = false
        `Corps principal\Congé arête.6\Activité` = false
        `Corps principal\Plan.2\Activité` = false
        `Corps principal\Extrusion.1\Activité` =true
        `Corps principal\Symétrie.1\Activité` = true
        `Corps principal\Congé arête.9\Activité` =true
}
```

Figure 1: Code for American/European case

By double-clicking the parameter directly in the tree, the user can instantly toggle between the two standardized socket models, proving the efficiency of the parameterization :

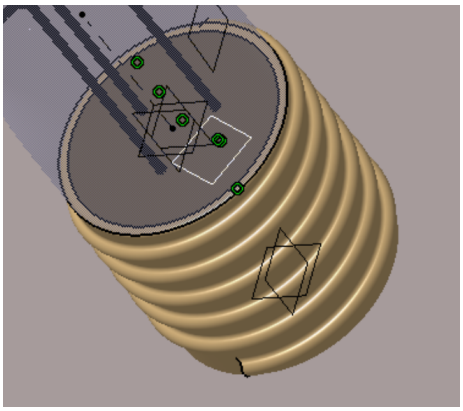

Figure 2: American Socket configuration

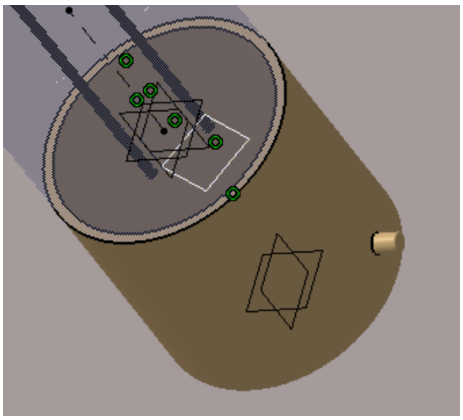

Figure 3: European Socket configuration

2 Exercise 2: Assembly Automation and Control

The second exercise expands KBE concepts from the Part level to the broader Assembly (Product) level, introducing mathematical formulas, safety checks, and data-driven design tables.

2.1 Assembly User Parameters

Global drivers were defined at the top assembly level to control the entire product architecture dynamically:

- **Bulb_Diameter** & **Bulb_Height**: Length type parameters governing the glass envelope's main proportions.
- **Socket_Type**: String type parameter (American/European).
- **Cost**: Real type parameter used for economic estimation.
- **Number_of_supports**: Integer type parameter defining the internal structure.

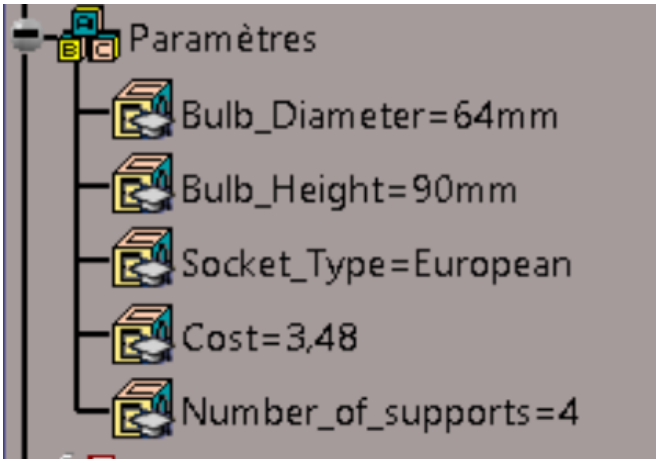

Figure 4: Parameters defined for Exercise 2

2.2 Rules for Supports and Cost

The internal geometry and economic estimations are driven by two main rules, showcasing CATIA's ability to handle both physical and non-physical data.

Support Logic: The `Number_of_supports` is determined by a conditional rule that checks if the `Bulb_Diameter` exceeds 60mm. If true, the volume requires 4 supports; otherwise, 3 supports are deemed sufficient.

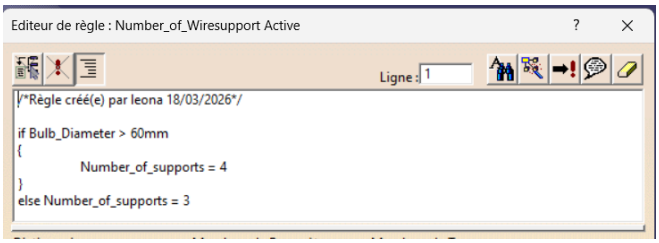

Figure 5: Bulb diameter rule for wire supports

Cost Calculation: The `Cost` parameter computes the estimated price based on the material volume (derived from Height and Diameter) and the manufacturing complexity associated with the chosen socket type.

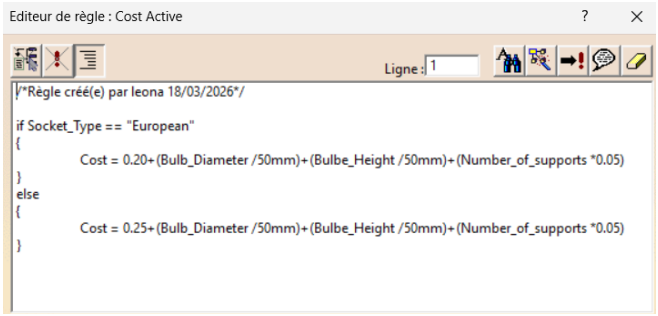

Figure 6: Cost calculation rule

2.3 Geometric Relations (Formulas)

To physically alter the 3D model, relations were established to link the Assembly parameters to the Part's internal constraints. For instance, the Offset distance of the bulb's main sketch is directly tied to the `Bulb_Height` parameter via a formula. This ensures the 3D updates automatically whenever the numerical value changes.

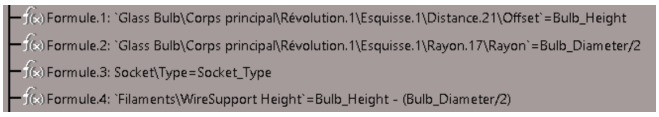

Figure 7: Formulas linking parameters to geometry

We also verify that the relations trigger the correct updates within the sub-components :

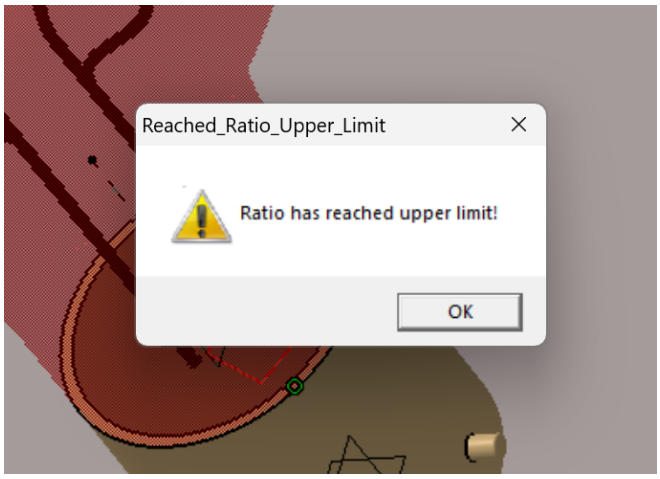

Figure 8: Rule triggering sub-component updates

2.4 Knowledge Checks for Quality Assurance

To prevent the generation of non-manufacturable or unviable proportions, **Checks** were implemented to continuously monitor the Bulb Ratio (H/D):

- **Upper Limit**: Triggers a Warning if $H/D < 1.7$ is violated.
- **Lower Limit**: Triggers a Warning if $H/D > 1.125$ is violated.

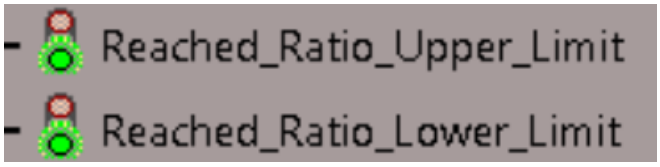

Figure 9: Checks configuration in the tree

If the designer inputs values outside this acceptable envelope, the check indicator immediately turns red and a specific warning message is displayed on the screen, alerting the user of the design flaw.

2.5 Design Table Integration

To streamline the design process, a **Design Table** (*Bulb_ Catalog*) was created directly from the current model parameters. This table was exported to Excel, allowing us to build a catalog of standard industrial configurations (e.g., 48mm × 68mm American, 64mm × 90mm European). Changing the configuration row instantly updates all connected geometry, rules, and checks.

Furthermore, we managed to analyze our boundary conditions using the **Knowledge Inspector**. This tool allowed us to run "What-If" scenarios to

see the impacts of a parameter change before applying it.

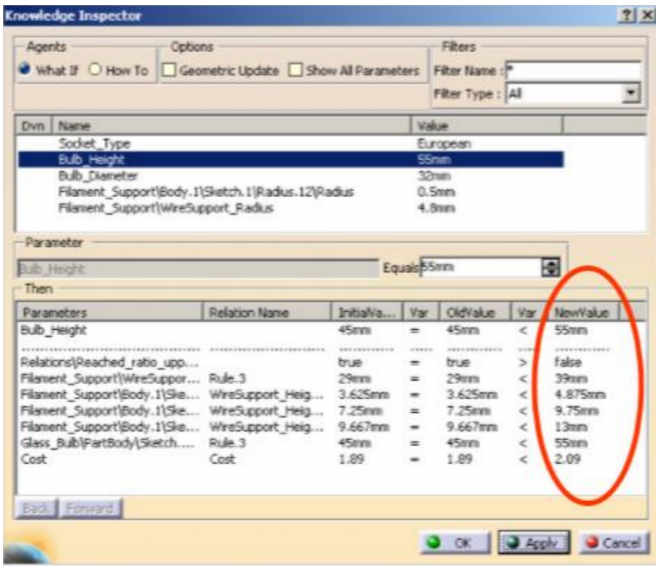

Figure 10: Knowledge Inspector analyzing a 'What-If' scenario

Bulb_Diameter (mm)	Bulb_Height (mm)	Socket_Type
48	68	American
48	80	American
48	90	American
64	90	European
64	64	European

Figure 11: Excel Design Table with standard configurations

2.6 Final Verification and Validation

As a final validation step, we tested the robustness of our rules and checks by loading two different configurations from the catalog:

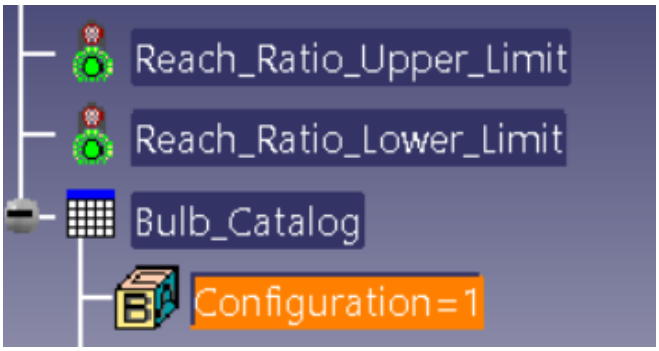

Figure 12: Valid configuration: Both checks pass (Green)

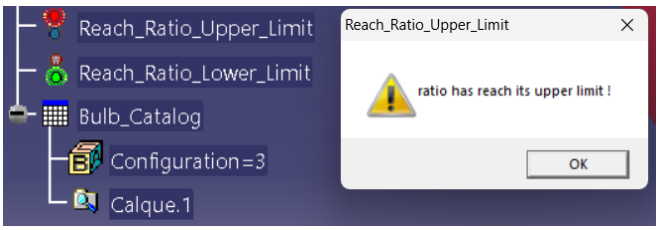

Figure 13: Invalid configuration: Upper limit check fails (Red)

Conclusion

This project successfully demonstrated the core purpose and power of **Knowledge-Based Engineering** in CATIA V5. By completing both exercises, a clear transition from static drafting to dynamic, rule-based modeling was achieved. The resulting smart light bulb assembly is highly adaptive and capable of:

- **Instant Updates**: Driving complex geometry through simple global parameters and formulas.
- **Embedded Logic**: Automating design and engineering decisions via Rules (e.g., adapting wire supports and calculating cost dynamically).
- **Design Integrity**: Ensuring technical compliance and preventing manufacturability issues through real-time Checks.
- **Standardization**: Managing multiple industrial iterations efficiently via integrated Design Tables.

In conclusion, implementing these rules drastically reduces the potential for human error and significantly accelerates the design cycle, highlighting exactly why KBE is a standard practice in modern aeronautical engineering.